Problem 29.1: Higgs Production at LEP

Source: M. D. Schwartz, Quantum Field Theory and the Standard Model, Problem 29.1.

The dominant production mechanism for Higgs bosons at LEP was $e^+e^- \to ZH$. Calculate the total cross section for this process at tree-level in the Standard Model. How many 100 GeV Higgs bosons would there have been when LEP ran at 206 GeV?

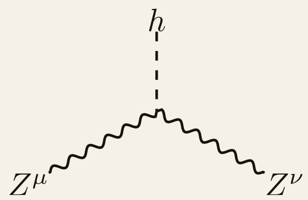
Solution. In this question, the relevant sector of the electroweak Lagrangian at tree-level is given by

$$\mathcal{L} = -\frac{1}{2}h(\Box + m_h^2)h + \bar{\psi}(i\cancel{\partial} - m_e)\psi + \frac{1}{v}m_Z^2 Z_\mu Z^\mu h + \frac{e}{\sin\theta_w\cos\theta_w}Z_\mu(J^{3\mu} - \sin^2\theta_w J^{\text{EM}\mu}), \tag{1.1}$$

where $v = 2\frac{m_W}{e}\sin\theta_w$, $J_\mu^{\text{EM}} = -\bar{\psi}\gamma_\mu\psi$ and

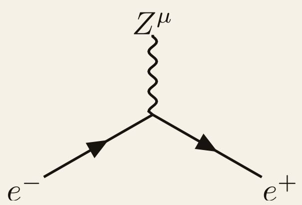
$$J_\mu^3 = \bar{\psi}^L\gamma_\mu T^3\psi^L = \bar{\psi}P_R\gamma_\mu T^3 P_L\psi = \bar{\psi}\gamma_\mu P_L T^3 P_L\psi = \bar{\psi}\gamma_\mu T^3 P_L\psi = -\frac{1}{2}\bar{\psi}\gamma_\mu P_L\psi, \tag{1.2}$$

where we have used $[P_L, T^3] = 0$, $T^3\psi^L = -\frac{1}{2}\psi^L$ and $P_L^2 = P_L$. Therefore, the ZZh interaction is given by



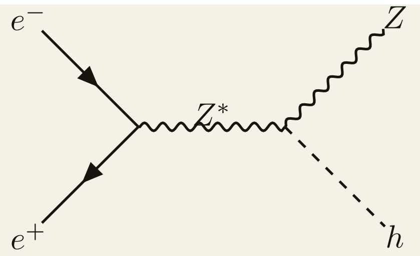
$$= i\frac{e}{\sin\theta_w\cos^2\theta_w}m_W g^{\mu\nu}. \tag{1.3}$$

The eeZ interaction is given by



$$= i\frac{e}{\sin\theta_w\cos\theta_w}\gamma^\mu\left(-\frac{1-\gamma^5}{4} + \sin^2\theta_w\right). \tag{1.4}$$

The only diagram we are concerned with at tree-level is



$$= i\mathcal{M}. \tag{1.5}$$

We can read off from the Feynman rules we just wrote down that

$$\begin{aligned} i\mathcal{M} &= \left(i\frac{e}{\sin\theta_w\cos\theta_w}\right)\left(i\frac{em_W}{\sin\theta_w\cos^2\theta_w}\right) \\ &\quad\times\left[\bar{v}(p_2)\gamma^\mu\left(-\frac{1-\gamma^5}{4}+\sin^2\theta_w\right)u(p_1)\right]\frac{-ig_{\mu\nu}}{s-m_Z^2}g^{\nu\rho}\epsilon^*_\rho(p_3) \\ &= \frac{ie^2m_W}{\sin^2\theta_w\cos^3\theta_w}\frac{1}{s-m_Z^2}\epsilon^*_\mu(p_3)\left[\bar{v}(p_2)\gamma^\mu\left(-\frac{1-\gamma^5}{4}+\sin^2\theta_w\right)u(p_1)\right]. \end{aligned} \tag{1.6}$$

Here the longitudinal part of the internal Z propagator vanishes by current conservation when $m_e = 0$. Taking the modulus square and summing over the external spins and polarisations, we find

$$\begin{aligned} \sum_{\text{spins/pols.}}|\mathcal{M}|^2 &= \left(\frac{e^2m_W}{\sin^2\theta_w\cos^3\theta_w}\right)^2\frac{1}{(s-m_Z^2)^2}\sum_{\text{pols.}}\epsilon^*_\mu(p_3)\epsilon_\nu(p_3) \\ &\quad\times\operatorname{Tr}\left[\not{p}_2\gamma^\mu\left(-\frac{1-\gamma^5}{4}+\sin^2\theta_w\right)\not{p}_1\gamma^\nu\left(-\frac{1-\gamma^5}{4}+\sin^2\theta_w\right)\right], \end{aligned} \tag{1.7}$$

where we have taken $m_e = 0$. The symmetric part of the trace over spinors is

$$\left(4\sin^4\theta_w - 2\sin^2\theta_w + \frac{1}{2}\right)\left(p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - p_1\cdot p_2 g^{\mu\nu}\right). \tag{1.8}$$

The antisymmetric part does not contribute when contracted with the polarisation sum, which is given by

$$\sum_{\text{pols.}}\epsilon^*_\mu(p_3)\epsilon_\nu(p_3) = -g_{\mu\nu} + \frac{p_{3\mu}p_{3\nu}}{m_Z^2}. \tag{1.9}$$

Putting them together, we find

$$\begin{aligned} \sum_{\text{spins/pols.}}|\mathcal{M}|^2 &= \left(\frac{e^2m_W}{\sin^2\theta_w\cos^3\theta_w}\right)^2\frac{4\sin^4\theta_w - 2\sin^2\theta_w + \frac{1}{2}}{(s-m_Z^2)^2} \\ &\quad\times\left(2p_1\cdot p_2 + \frac{2(p_1\cdot p_3)(p_2\cdot p_3)}{m_Z^2} - \frac{(p_1\cdot p_2)p_3^2}{m_Z^2}\right). \end{aligned} \tag{1.10}$$

To compute the cross-section, we need to average over the initial spins, which is to multiply by a factor of $1/4$. In the centre-of-mass frame, we have

$$\begin{gathered} p_3^2 = m_Z^2, \qquad p_1\cdot p_2 = \frac{s}{2}, \qquad |\mathbf{k}| = \frac{\sqrt{s}}{2}, \\ (p_1\cdot p_3)(p_2\cdot p_3) = \frac{s}{4}\left(E_Z^2 - |\mathbf{p}|^2\cos^2\theta\right), \qquad E_Z = \sqrt{|\mathbf{p}|^2 + m_Z^2}, \end{gathered} \tag{1.11}$$

where $\mathbf{k}$ is the momentum of the incoming electron and positron and $\mathbf{p}$ is the momentum of the outgoing Z and Higgs bosons. Hence,

$$\begin{aligned}\sum_{\text{spins/pols.}} |\mathcal{M}|^2 = \left(\frac{e^2 m_W}{\sin^2\theta_w \cos^3\theta_w}\right)^2 \frac{s\left(4\sin^4\theta_w - 2\sin^2\theta_w + \frac{1}{2}\right)}{(s-m_Z^2)^2} \\ \times \left[1 + \frac{\left[s-(m_Z+m_h)^2\right]\left[s-(m_Z-m_h)^2\right]}{8sm_Z^2}\sin^2\theta\right].\end{aligned} \tag{1.12}$$

The magnitude of the outgoing three-momentum is

$$|\mathbf{p}| = \frac{\sqrt{[s-(m_Z+m_h)^2]\,[s-(m_Z-m_h)^2]}}{2\sqrt{s}}, \tag{1.13}$$

and therefore

$$\frac{|\mathbf{p}|}{|\mathbf{k}|} = \frac{\sqrt{[s-(m_Z+m_h)^2]\,[s-(m_Z-m_h)^2]}}{s}. \tag{1.14}$$

It follows that

$$\begin{aligned}\frac{d\sigma}{d\Omega} &= \frac{1}{64\pi^2 s}\frac{|\mathbf{p}|}{|\mathbf{k}|}\frac{1}{4}\sum_{\text{spins/pols.}}|\mathcal{M}|^2 \\ &= \frac{\alpha^2 m_W^2\left(4\sin^4\theta_w - 2\sin^2\theta_w + \frac{1}{2}\right)}{16\sin^4\theta_w\cos^6\theta_w(s-m_Z^2)^2}\,\frac{\sqrt{[s-(m_Z+m_h)^2]\,[s-(m_Z-m_h)^2]}}{s} \\ &\quad\times\left[1+\frac{\left[s-(m_Z+m_h)^2\right]\left[s-(m_Z-m_h)^2\right]}{8sm_Z^2}\sin^2\theta\right].\end{aligned} \tag{1.15}$$

Using $\int d\Omega = 4\pi$ and $\int d\Omega\,\sin^2\theta = 8\pi/3$, the total cross section is

$$\begin{aligned}\sigma(e^+e^-\to Zh) = \frac{\pi\alpha^2 m_W^2\left(4\sin^4\theta_w - 2\sin^2\theta_w + \frac{1}{2}\right)}{4\sin^4\theta_w\cos^6\theta_w(s-m_Z^2)^2}\,\frac{\sqrt{[s-(m_Z+m_h)^2]\,[s-(m_Z-m_h)^2]}}{s} \\ \times\left[1+\frac{\left[s-(m_Z+m_h)^2\right]\left[s-(m_Z-m_h)^2\right]}{12sm_Z^2}\right].\end{aligned} \tag{1.16}$$

For the numerical estimate, we take

$$\sqrt{s} = 206\,\text{GeV}, \qquad m_h = 100\,\text{GeV}, \qquad m_Z = 91.2\,\text{GeV}, \qquad m_W = 80.4\,\text{GeV}, \tag{1.17}$$

together with

$$\sin^2\theta_w = 0.231, \qquad \alpha = \frac{1}{128}. \tag{1.18}$$

Substituting these values into the total cross section gives

$$\sigma(e^+e^-\to Zh) \simeq 1.09\times 10^{-9}\,\text{GeV}^{-2}. \tag{1.19}$$

Using

$$1\,\text{GeV}^{-2} = 3.894\times 10^8\,\text{pb}, \tag{1.20}$$

this becomes

$$\sigma(e^+e^- \to Zh) \simeq 0.42\,\mathrm{pb}. \tag{1.21}$$

The four LEP experiments collected an integrated luminosity of approximately $536\,\mathrm{pb}^{-1}$ at centre-of-mass energies of $206\,\mathrm{GeV}$ or higher. The expected number of produced Higgs bosons is therefore

$$N = \sigma L_{\mathrm{int}} \simeq (0.42\,\mathrm{pb})(536\,\mathrm{pb}^{-1}) \simeq 2.3 \times 10^2. \tag{1.22}$$

Thus, approximately

$$\boxed{N \simeq 230} \tag{1.23}$$

$100\,\mathrm{GeV}$ Higgs bosons would have been produced.